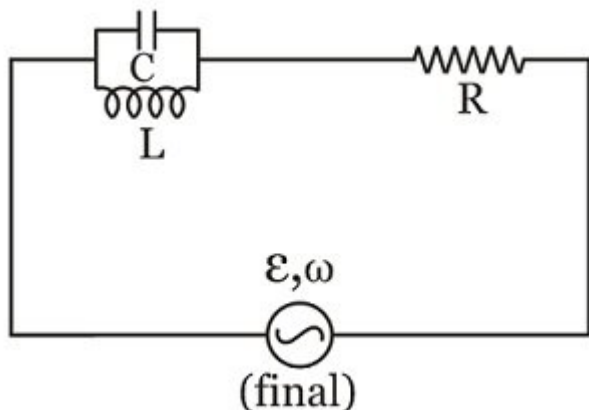


Q1

The current flowing through the resistor in a series RLC *a. c.* circuit, is $I = \frac{\varepsilon}{R}$. Now the inductor and capacitor are connected in parallel and joined in series with the resistor as shown in figure. The current in the circuit is now.
(symbols have their usual meaning)



- (1) Equal to I
- (2) More than I
- (3) Less than I
- (4) Zero

Q2

Pick the correct statements(s):

- (1) The rms translational speed for all ideal-gases molecules at the same temperature is not the same but it depends on the mass
- (2) Each particle in a gas has average translational kinetic energy and the equation $\frac{1}{2} m v_{rms}^2 = \frac{3}{2} kT$ establishes the relationship between the translational kinetic energy of a particle and temperature of an ideal gas
- (3) Temperature of an ideal gas is doubled from 100°C to 200°C . The average kinetic energy of each particle is also doubled
- (4) It is possible for both the pressure and volume of a monoatomic ideal gas to change simultaneously without causing the internal energy of the gas to change

Q3

The de-broglie wavelength of 2 elementary particles in centre of mass frame is same and is equal to $5/8$ times de-broglie wavelength of the lighter particle in lab frame and $5/2$ times the de-broglie wavelength of the heavier particle in lab frame. Both particles have either parallel or anti parallel velocities. The possible combinations of particles are :

Questions with Answer Keys

MathonGo

(1) ${}^6_{13}\text{C}$ and ${}^6_{12}\text{C}$

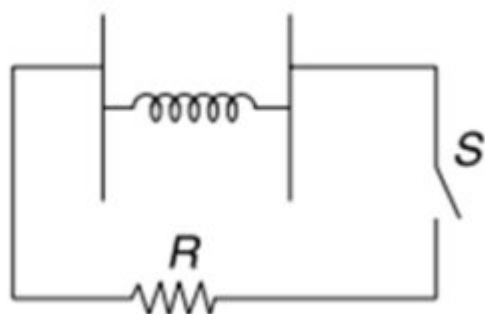
(2) ${}^6_{13}\text{C}$ and ${}^2_4\text{He}$

(3) ${}^1_1\text{H}$ and ${}^2_4\text{He}$

(4) ${}^3_7\text{Li}$ and ${}^6_{13}\text{C}$

Q4

A parallel plate capacitor has its two plates connected to an ideal spring of force constant K . Relaxed length of spring is L and it is made of non-conducting material. The area of each plate is A . The capacitor has a charge q_0 on it. To discharge the capacitor through the resistance R , switch S is closed. Assume that there is no friction and the plates always remains parallel to each other.



(1) If the time constant of the circuit is very large and discharge process is very slow, heat dissipated in the resistance is $\frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{4\epsilon_0^2 A^2 k}$

(2) If the time constant of the circuit is very large and discharge process is very slow, heat dissipated in the resistance is $\frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{8\epsilon_0^2 A^2 k}$

(3) If the time constant of the circuit is very small and discharge process is almost instantaneous, heat dissipated in the resistance is $\frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{4\epsilon_0^2 A^2 k}$

(4) If the time constant of the circuit is very small and discharge process is almost instantaneous, heat dissipated in the resistance is $\frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{8\epsilon_0^2 A^2 k}$

Q5

A particle of mass m moves on the x -axis as follows: It starts from rest at $t = 0$ from the point $x = 0$, and comes to rest at $t = 1$ at the point $x = 1$. No other information is available about its motion at intermediate times ($0 < t < 1$). If α denotes the instantaneous acceleration of the particle, then:

(1) α cannot remain positive for all t in the interval $0 \leq t \leq 1$

(2) $|\alpha|$ cannot exceed 2 at any point in its path

Questions with Answer Keys

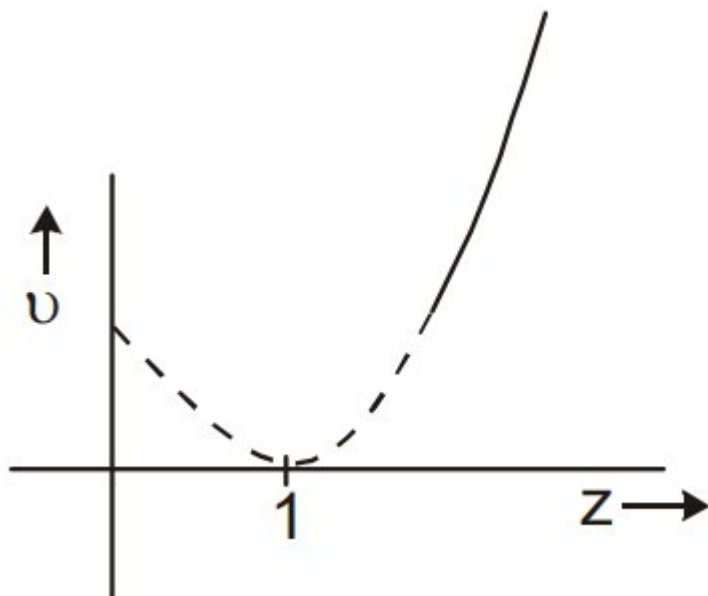
MathonGo

(3) $|\alpha|$ must be ≥ 4 at some point of points in its path

(4) α must change sign during the motion, but no other assertion can be made with the information given

Q6

Figure shows the variation of frequency of a characteristic X-ray and atomic number.



(1) The characteristic X-ray is K_{α}

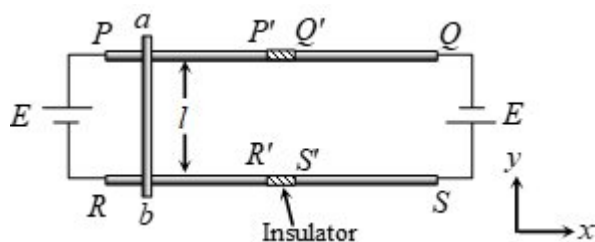
(2) The characteristic X-ray is K_{β}

(3) The energy of photon emitted when this X-ray is emitted by a metal having $z = 101$ is 204 keV

(4) The energy of photon emitted when this X-ray is emitted by a metal having $z = 101$ is 102 keV

Q7

System shows a slidable conductor ab of mass m which can move over frictionless and resistanceless rails $PP'Q'Q$ and $RR'S'S$ in horizontal plane. PP' , QQ' and RR' , $S'S$ are separated by a very small insulator as shown in the figure. At $t = 0$ a magnetic field $-B\hat{k}$ Tesla is applied everywhere and conductor ab is released from rest. Choose the correct statement (s). Neglecting back emf in the conductor ab .



(1) Motion of conductor ab is S.H.M.

Questions with Answer Keys

MathonGo

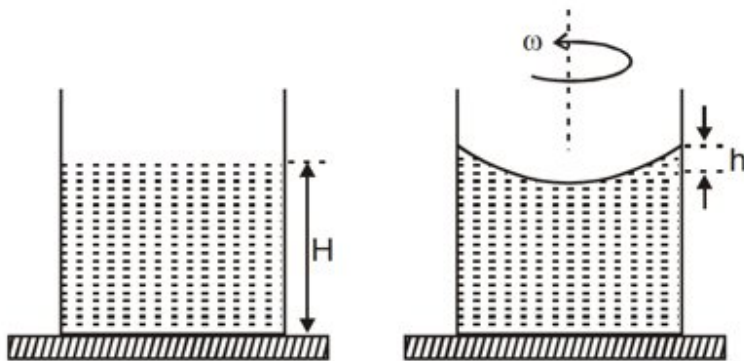
(2) Time taken by conductor ab to reach back the initial position must be $\sqrt{\frac{32ymR}{BlE}}$, where R is resistance of conductor ab and $PP' = QQ' = y$.

(3) If the battery on the right hand side is removed and a conductor identical to ab is kept on the rails near the end Q and S and ab is released from the position shown the new conductor will tend to move in the direction opposite to the motion of ab due to electromagnetic induction.

(4) To keep the new conductor at its position the minimum coefficient of friction between rails and new conductor must be $\mu = \frac{BEI}{m^2g}$.

Q8

A beaker of radius r is filled with water (refractive index $\frac{4}{3}$) up to a height H as shown in the figure on the left. The beaker is kept on a horizontal table rotating with angular speed ω . This makes the water surface curved so that the difference in the height of water level at the center and at the circumference of the beaker is h ($h \ll H, h \ll r$), as shown in the figure on the right. Take this surface to be approximately spherical with a radius of curvature R . Which of the following is/are correct ? (g is the acceleration due to gravity)



(1) $R = \frac{h^2 + r^2}{2h}$

(2) $R = \frac{3r^2}{2h}$

(3) Apparent depth of the bottom of the beaker is close to $\frac{3H}{2} \left(1 + \frac{\omega^2 H}{2g}\right)^{-1}$

(4) Apparent depth of the bottom of the beaker is close to $\frac{3H}{4} \left(1 + \frac{\omega^2 H}{4g}\right)^{-1}$

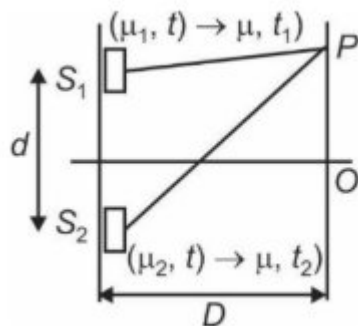
Q9

In a YDSE two thin transparent sheets are used in front of the slits S_1 and S_2 having refractive indices $\mu_1 = 1.6$ and $\mu_2 = 1.4$ respectively. If both sheets have thickness ' t ', the central maximum is observed at a distance of 5 mm from centre O . Now the sheets are replaced by two sheets of same material of refractive index $\frac{\mu_1 + \mu_2}{2}$ but having thickness t_1 and t_2 such that ' $t' = \frac{t_1 + t_2}{2}$ '. After replacement, now central maximum is observed at distance of 8 mm from centre O on the same side as before. Find the thickness t_1 (in μm).

Questions with Answer Keys

MathonGo

[Given $d = 1 \text{ mm}$, $D = 1 \text{ m}$] (mark the answer in $\mu \text{ m}$ to nearest integer)

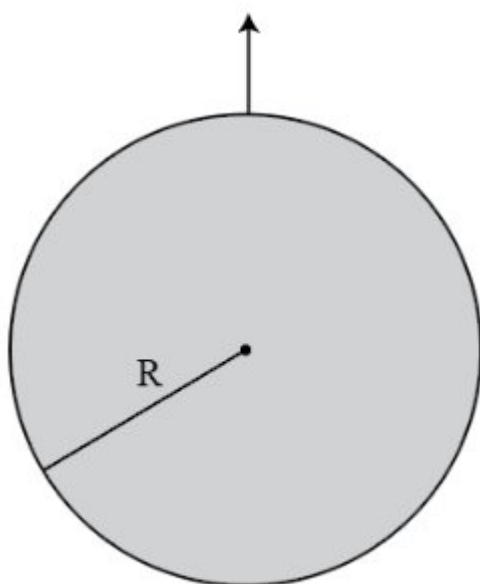


Q10

A container with 1 kg of water in it is kept in sunlight, which causes the water to get warmer than the surroundings. The average energy per unit time per unit area received due to the sunlight is 700 W m^{-2} and it is absorbed by the water over an effective area of 0.05 m^2 . Assuming that the heat loss from the water to the surroundings is governed by Newton's law of cooling, the difference (in $^{\circ}\text{C}$) in the temperature of water and the surroundings after a long time will be (Ignore effect of the container, and take constant for Newton's law of cooling $= 0.001 \text{ s}^{-1}$, Heat capacity of water $= 4200 \text{ J kg}^{-1} \text{ K}^{-1}$)

Q11

A small body of mass ' m ' is projected up from the surface of given planet as shown in the figure. If small body has velocity which is just sufficient to get out from the gravity of planet then time taken by small body to reach at height of $3R$ from the surface of planet is $K \sqrt{\frac{2R^3}{GM}}$. Find the value of K .



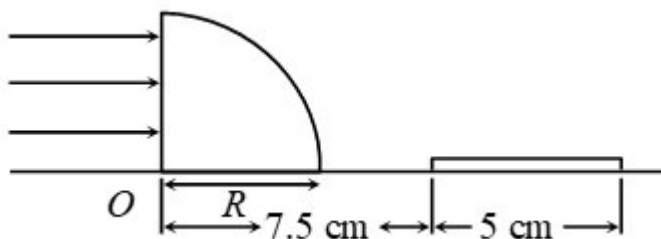
Planet of mass M
and radius R

Questions with Answer Keys

MathonGo

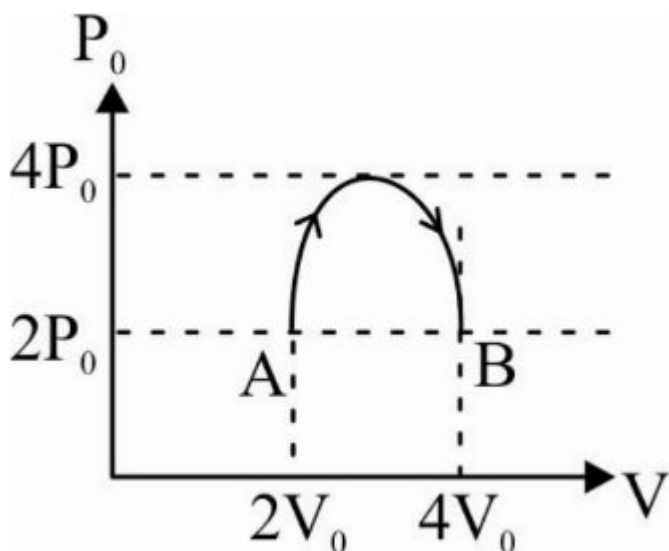
Q12

A glass ($\mu = 2$) prism is in the shape of a quarter cylinder of length L and radius $R = 5 \text{ cm}$, lying on a horizontal table. A uniform horizontal light beam fall on its vertical plane surfaces, as shown in figure. If a plane paper of width 5 cm and length L is placed at a distance 7.5 cm from O , then the fraction of light (in percentage) falling on the prism that will fall come out of prism is _____



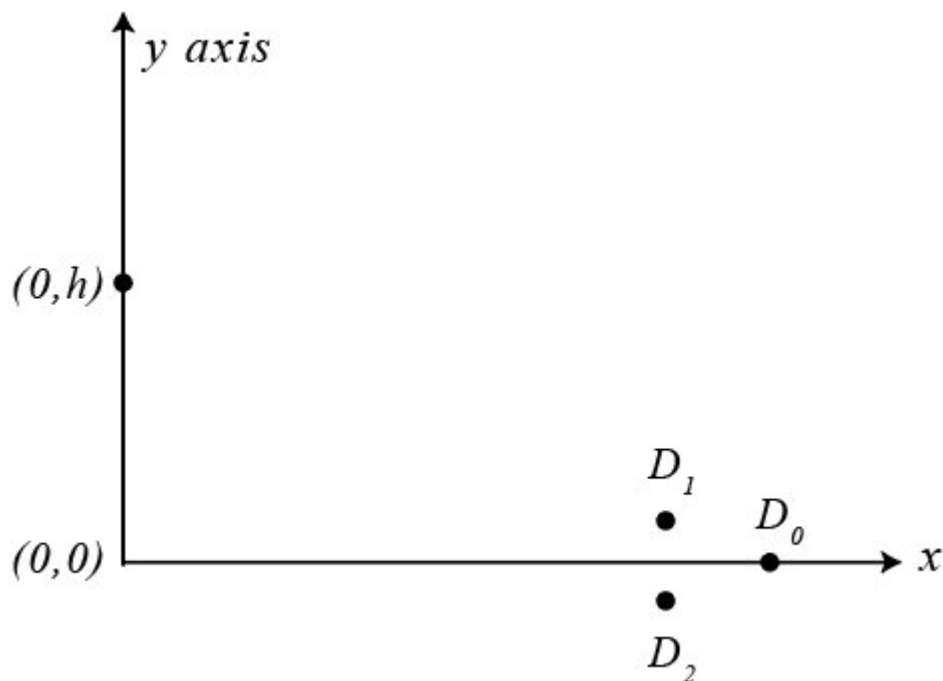
Q13

One mole of an ideal monoatomic gas is taken through a thermodynamic process shown in the $P - V$ diagram. The heat supplied to the system in this process is $K \times (\pi + 10)P_0 V_0$. Determine the value of K .



Q14

There is a point source of sound placed at $(0, h)$ as shown in figure. The detectors D_1 and D_2 are placed at positions $(D, d/2)$ and $(D, -d/2)$ respectively. Take $h \ll D$. The source emitted a sound pulse at a certain time. Assuming velocity of sound in the surrounding medium as V . (Assume that the waves reaching the detectors D_1, D_2 are plane progressive waves)

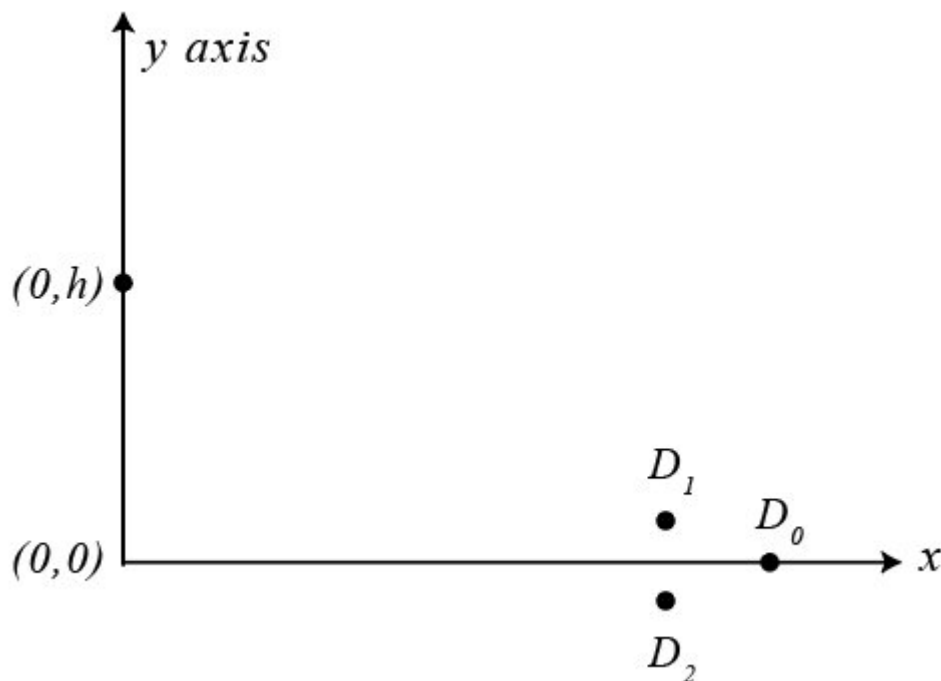


The time gap between the recordings made by the detectors will approximately be

- (1) $\frac{hd}{2DV}$
- (2) $\frac{2hd}{DV}$
- (3) $\frac{hd}{DV}$
- (4) $\frac{4hd}{DV}$

Q15

There is a point source of sound placed at $(0, h)$ as shown in figure. The detectors D_1 and D_2 are placed at positions $(D, d/2)$ and $(D, -d/2)$ respectively. Take $h \ll D$. The source emitted a sound pulse at a certain time. Assuming velocity of sound in the surrounding medium as V . (Assume that the waves reaching the detectors D_1, D_2 are plane progressive waves)

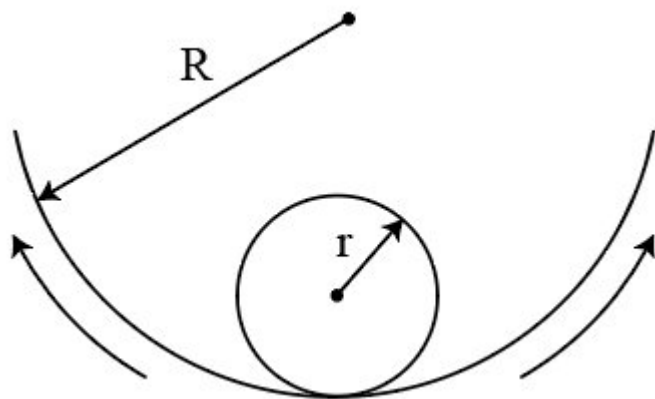


If the source emits continuous waves, and the pressure recorded by the two detectors are superposed at every instant in detector D_0 (which is equidistant from D_1 & D_2) the resultant pressure amplitude will be maximum if the minimum frequency of the source is

- (1) $\frac{VD}{2dh}$
- (2) $\frac{2VD}{dh}$
- (3) $\frac{3VD}{2dh}$
- (4) $\frac{VD}{dh}$

Q16

Paragraph: Consider a sphere filled with liquid inside rolling back and forth at the bottom of a spherical bowl. That is, the sphere is periodically changing its translational and rotational direction. Assume that the rigid thin spherical shell of radius r and mass m is fully filled with some liquid substance of mass M . Liquid has such a unique property that usually it behaves like an ideal liquid (i.e. without any viscosity), while in response to some special external influence (such as electric field or sudden cold climate) it transits to solid state immediately with the same volume; and once the applied influence removed, the liquid state recovers immediately. Besides, this influence does not give rise to any force or torque exerting on the sphere. This liquid-filled spherical shell (for convenience, called 'the sphere' hereafter) is supposed to roll back and forth at the bottom of a spherical bowl of radius R without any relative slipping, as shown in the figure. Assume the sphere moves only in the vertical plane.

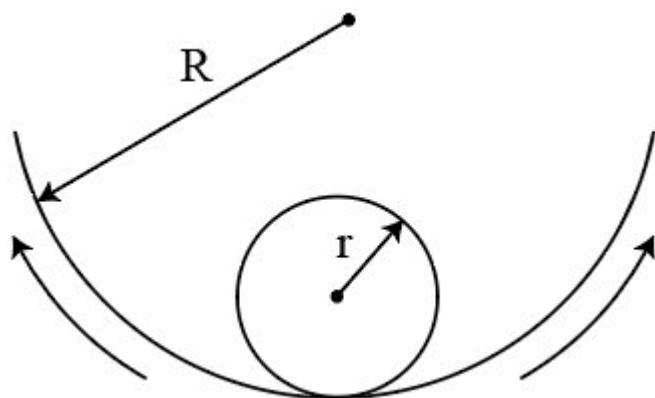


Question: If the transition of spherical shell of liquid to solid occurs when the shell is at its extreme, Which of the following statements is CORRECT?

- (1) The angular amplitude remains the same, the oscillation energy remains the same and the speed at the bottom increases.
- (2) The angular amplitude remains the same, the oscillation energy remains the same and the speed at the bottom decreases.
- (3) The angular amplitude decreases, the oscillation energy increases and the speed at the bottom decreases.
- (4) The angular amplitude, the oscillation energy and the speed at the bottom all remain same.

Q17

Paragraph: Consider a sphere filled with liquid inside rolling back and forth at the bottom of a spherical bowl. That is, the sphere is periodically changing its translational and rotational direction. Assume that the rigid thin spherical shell of radius r and mass m is fully filled with some liquid substance of mass M . Liquid has such a unique property that usually it behaves like an ideal liquid (i.e. without any viscosity), while in response to some special external influence (such as electric field or sudden cold climate) it transits to solid state immediately with the same volume; and once the applied influence removed, the liquid state recovers immediately. Besides, this influence does not give rise to any force or torque exerting on the sphere. This liquid-filled spherical shell (for convenience, called 'the sphere' hereafter) is supposed to roll back and forth at the bottom of a spherical bowl of radius R without any relative slipping, as shown in the figure. Assume the sphere moves only in the vertical plane.



Questions with Answer Keys

MathonGo

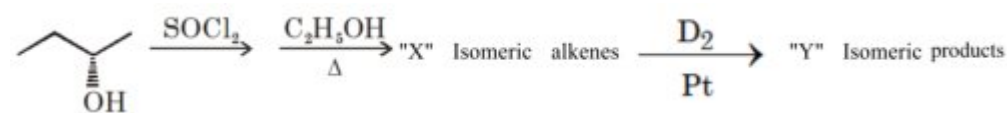
Question: If the transition of spherical shell of liquid to solid occurs when the shell is at its mean position, Which of the following statements is CORRECT?

- (1) The angular amplitude decreases, the oscillation energy decreases and the speed at the bottom decreases.
- (2) The angular amplitude remains the same, the oscillation energy decreases and the speed at the bottom decreases.
- (3) The angular amplitude remains same, the oscillation energy remains same and the speed at the bottom remains same.
- (4) The angular amplitude decreases, the oscillation energy increases and the speed at the bottom decreases.

Q18

Which of the following is a correct statement about s-block compounds?

- (1) Stability of metal chlorides decreases down the alkali metal group.
- (2) Lattice energy of M_2O_2 is greater than that of M_2O .
- (3) Stability of peroxide increases with the increase in size of alkali metal cation.
- (4) The water of crystallization is greater in alkali metal salts than in alkaline earth metal salts.

Q19

(X + Y) is :

- (1) 3
- (2) 2
- (3) 4
- (4) 8

Q20

In self reduction the reducing species is

Select the correct option for blank.

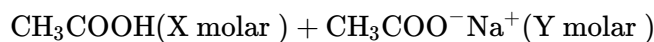
- (1) Cu^+
- (2) O^{-2}
- (3) S^{-2}

Questions with Answer Keys

(4) SiO_2

Q21

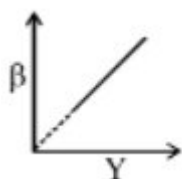
Select the correct statement(s) regarding buffer solution :



(1) At constant buffer concentration, buffer capacity increases to its maximum value if $\frac{Y}{X}$ approaches 1.

(2) At constant buffer concentration, variation of index of buffer with Y is parabolic in nature.

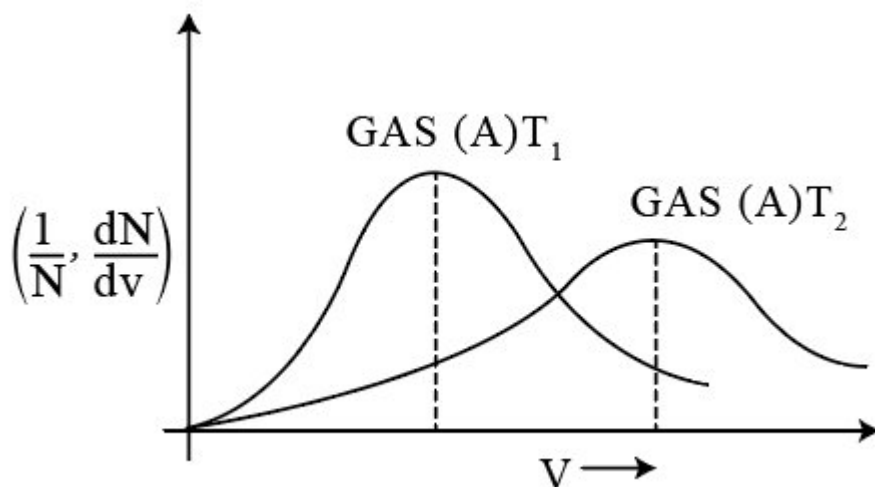
(3) Keeping $\frac{Y}{X} = 1$, β (buffer capacity) vs Y can be represented by



(4) On dilution, pH of the buffer solution remains constant whereas its buffer capacity increases.

Q22

According to Maxwell's distribution of molecular speeds, for the below graph drawn for two different samples of gases A and B at temperature T_1 and T_2 respectively, which of the following statements is/are INCORRECT :



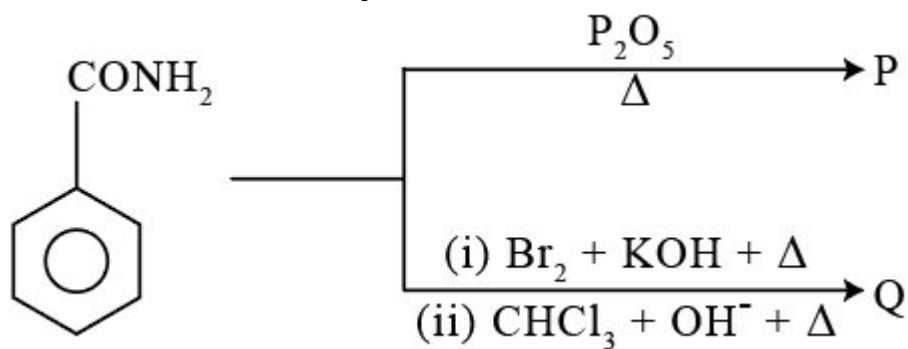
(1) If $T_1 = T_2$, then molecular mass of gas B (M_B) is greater than molecular mass of gas A (M_A).

(2) If molecular mass of gas A (M_A) is equal to molecular mass of gas B (M_B), then $T_1 > T_2$.

(3) If $T_1 < T_2$, then molecular mass of gas B (M_B) is necessarily less than molecular mass of gas A (M_A).

(4) If gas A is O_2 and gas B is N_2 , then considering them to be ideal gases, T_1 is necessarily less than T_2 .

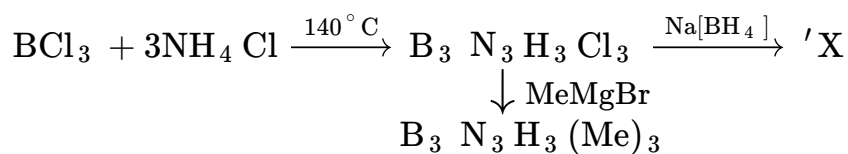
Q23



Which statement(s) is/are correct :

- (1) Reduced product of P and Q will be metamers to each other.
- (2) By dry distillation of hydrolysed products of P with $Ca(OH)_2$, gives benzophenone.
- (3) Hydrolysed product of Q, reacts with $NaNO_2 + HCl$ followed by reaction with phenol, gives orange red dye.
- (4) Electrophile involved in the formation of Q is dichlorocarbene.

Q24



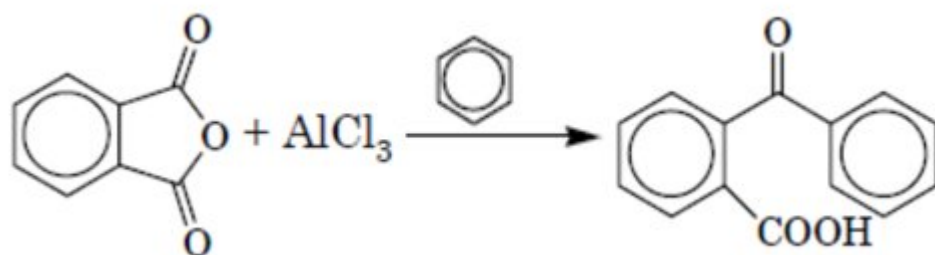
Which of the following statements are CORRECT :-

- (1) 'X' is known as inorganic graphite
- (2) 'X' on hydrolysis along with heating gives NH_3 , H_2 and $B(OH)_3$
- (3) 'X' is known as inorganic benzene
- (4) 'X' has $2p_\pi - 2p_\pi$ back bonding

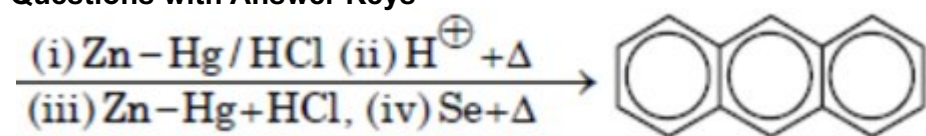
Q25

Which of the following reaction(s) give correct product as shown

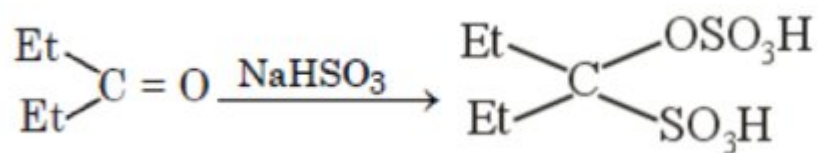
(1)



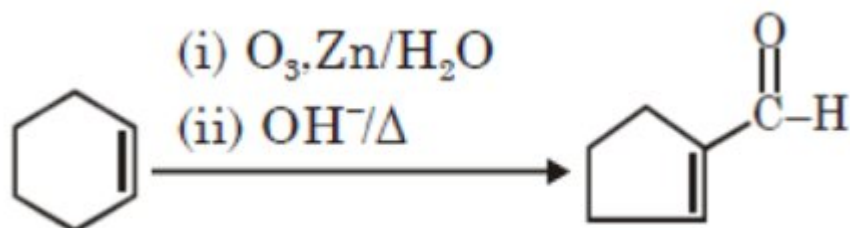
Questions with Answer Keys



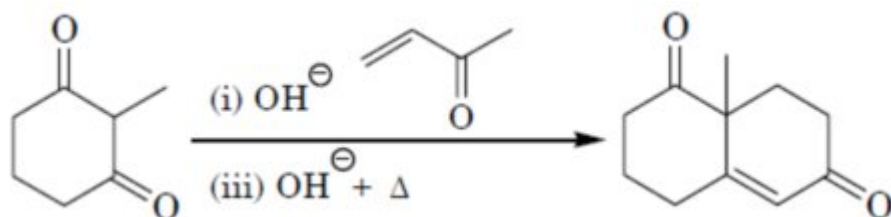
(2)



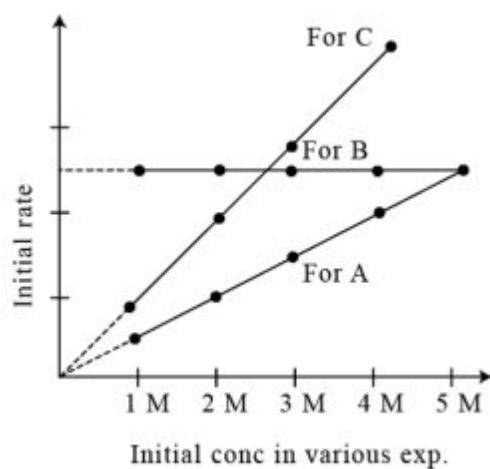
(3)



(4)



Q26



If three reactions :

Questions with Answer Keys

A $\longrightarrow$ Products Rate = $K_1[A]^X$

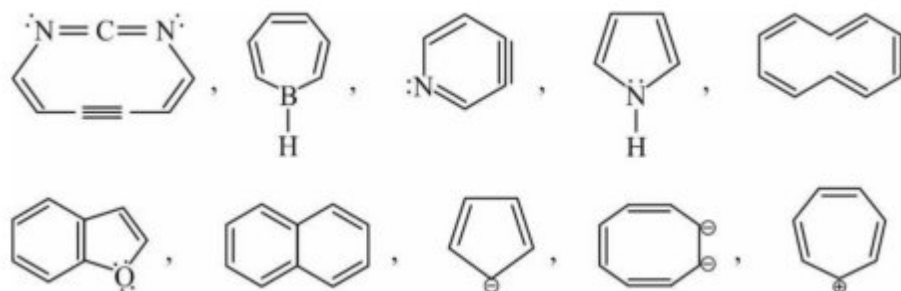
B $\longrightarrow$ Products Rate = $K_2[B]^Y$

C $\longrightarrow$ Products Rate = $K_3[C]^Z$

Value of $\left(\frac{X+Y}{4Z}\right)$ is?

Q27

Among the following, the number of aromatic compound(s) having only sp^2 hybridized atoms is



Q28

An acidified solution of potassium chromate was layered with an equal volume of amyl alcohol. When it was shaken after the addition of 1 mL of 3% H_2O_2 , a blue alcohol layer was obtained. The blue color is due to the formation of a chromium (VI) compound 'X'. What is the number of oxygen atoms bonded to chromium through only single bonds in a molecule of X?

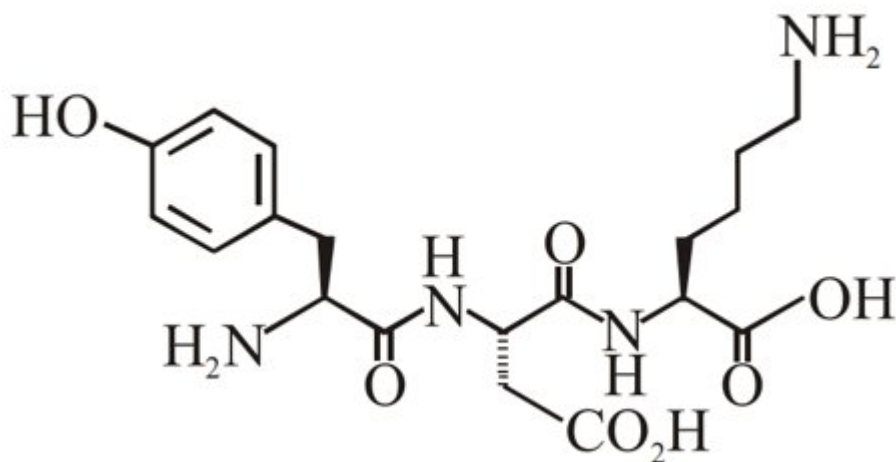
Q29

The sum of number of unpaired electrons in each of the following species is

B_2 , XeF , $[Fe(H_2O)_5(NO)]SO_4$, K_3CrO_8 , $Ba(O_2)_2$

Q30

The structure of a peptide is given below.



Questions with Answer Keys

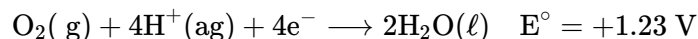
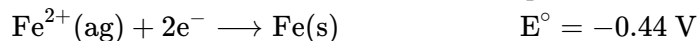
MathonGo

If the absolute values of the net charge of the peptide at $\text{pH} = 2$, $\text{pH} = 6$, and $\text{pH} = 11$ are $|z_1|$, $|z_2|$ and $|z_3|$, respectively, then what is $|z_1| + |z_2| + |z_3|$?

Q31

Paragraph:

The corrosion of iron is an electrochemical process that involves the standard reduction potential given here at 25°C .



Question:

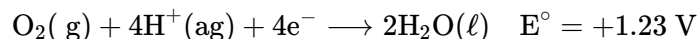
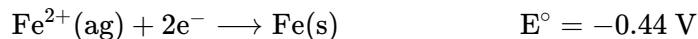
The voltage if the reaction occurs at $\text{pH} = 4.0$ but all other concentration are maintained as they were in the standard cell. (Given : $2.303RT/F = 0.06 \text{ V}$)

- (1) 1.43 V
- (2) 1.67 V
- (3) -0.44 V
- (4) none

Q32

Paragraph:

The corrosion of iron is an electrochemical process that involves the standard reduction potential given here at 25°C .



Question:

For the reaction $\text{Fe}(\text{OH})_2(\text{s}) + 2\text{e}^- \longrightarrow \text{Fe}(\text{s}) + 2\text{OH}^-(\text{aq})$

$E^\circ = -0.88 \text{ V}$. Use this information with one of the given standard potential to calculate the $\ln K_{\text{sp}}$ of $\text{Fe}(\text{OH})_2$ at 25°C . (Given : $2.303RT/F = 0.06 \text{ V}$)

- (1) -28.76
- (2) -33.77
- (3) -20.67
- (4) None of these

Q33

Paragraph:

An organic compound (X) is subjected to reductive ozonolysis to give a product (A) which on heating with dil. NaOH gives another product (B). Compound (B) on heating with sodium hypoiodite followed by acidification produces (C). Compound (C) on treatment with SOCl_2 and then with diazomethane followed by hydrolysis gives a product which on heating gives 1, 2-dimethyl cyclopentene.

Questions with Answer Keys

Now answer the following questions.

Question:

Organic compound (X) in the above passage is:

- (1) 1,2-dimethyl cyclopentene
- (2) 1,2-dimethyl hexene
- (3) 1,2-dimethyl cyclohexene
- (4) 2,5-dimethyl-3-hexene.

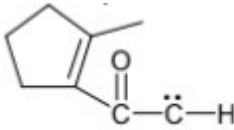
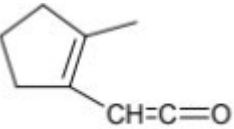
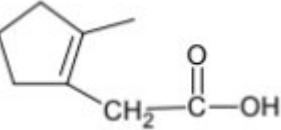
Q34**Paragraph:**

An organic compound (X) is subjected to reductive ozonolysis to give a product (A) which on heating with dil. NaOH gives another product (B). Compound (B) on heating with sodium hypoiodite followed by acidification produces (C). Compound (C) on treatment with SOCl_2 and then with diazomethane followed by hydrolysis gives a product which on heating gives 1, 2-dimethyl cyclopentene.

Now answer the following questions.

Question:

Intermediate species formed in the reaction of (C) with $\text{SOCl}_2/\text{CH}_2\text{N}_2/\text{H}_2\text{O}/\Delta$ is:

- (1) 
- (2) 
- (3) 
- (4) All of these

Q35

Let $\{a_n\}$ be a sequence such that $a_0 = 1, a_1 = 0, a_n = 3a_{n-1} - 2a_{n-2}$. The correct statement is:

- (1) $a_{45} = 2^{45}$
- (2) $a_{51} = 2^{25} - 2$
- (3) $a_{48} = 2(1 - 2^{47})$

Questions with Answer Keys

(4) $a_{49} = \sqrt{2} - \sqrt{2^{49}}$

Q36

Let $f(x) = \max\{a, b, c\}$, where

$$a = \lim_{n \rightarrow \infty} \lim_{\alpha \rightarrow 1^+} \frac{\alpha^n |\sin x| + \alpha^{-n} |\cos x|}{\alpha^n + \alpha^{-n}}$$

$$b = \lim_{n \rightarrow \infty} \lim_{\alpha \rightarrow 1^-} \frac{\alpha^n |\sin x| + \alpha^{-n} |\cos x|}{\alpha^n + \alpha^{-n}}$$

$$c = \lim_{n \rightarrow \infty} \frac{\pi}{4n} \left[1 + \cos \frac{\pi}{2n} + \cos \frac{2\pi}{2n} + \dots + \cos \frac{(n-1)\pi}{2n} \right]. \text{ Then}$$

Range of $f(x)$ is

(1) $[0, 1]$

(2) $\left[\frac{1}{2}, 1\right]$

(3) $\left[\frac{1}{\sqrt{2}}, 1\right]$

(4) $\left[\frac{1}{2}, 2\right]$

Q37

If $\Delta(x) = \begin{vmatrix} 3 & 3x & 3x^2 + 2a^2 \\ 3x & 3x^2 + 2a^2 & 3x^3 + 6a^2x \\ 3x^2 + 2a^2 & 3x^3 + 6a^2x & 3x^4 + 12a^2x^2 + 2a^4 \end{vmatrix}$, then which of the following options are always true for all

$a \in R$?

(1) $\Delta'(x) = 0$

(2) $\Delta(x)$ is not independent of x

(3) $\int_0^1 \Delta(x) dx = 16a^6$

(4) $y = \Delta(x)$ is the equation of a circle.

Q38

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors each of magnitude $\sqrt{3}$ and the angle between each pair of them is $\frac{\pi}{3}$. If

$\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = x\vec{a} + y\vec{b} + z\vec{c}$ where x, y, z are scalars, then

(1) volume of parallelepiped formed by $\vec{a}, \vec{b}, \vec{c}$ is $\frac{3\sqrt{3}}{\sqrt{2}}$

(2) the value of x is $\frac{\sqrt{3}}{\sqrt{2}}$

(3) $x^2 + y^2 + z^2 \leq \sqrt{21}$

Questions with Answer Keys

MathonGo

(4) Number of integral co-ordinates (x, y, z) satisfying $x^2 + y^2 + z^2 \leq 12$ is 125

Q39

Let $C_k = {}^nC_k$ for $0 \leq k \leq n$ and $A_k = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix}$ for $k \geq 1$ and

$A_1 A_2 + A_2 A_3 + A_3 A_4 + \dots + A_{n-1} A_n = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ then

(1) $a = b$

(2) $a + b = 2 ({}^{2n}C_{n-1} - n)$

(3) $a = 2b$

(4) $a + b = 0$

Q40

Consider a conic $C : y^2 = 4x$. Let $\triangle PQR$ be an equilateral triangle with side length k where P be any point on C , Q be the foot of perpendicular from P upon the directrix of C and R be the focus of C . A circle C_2 is inscribed in another conic $C_1 : y^2 = k(x + 1)$ which touches C_1 at the points where C_1 cuts the y -axis. C_3 is an ellipse whose auxiliary circle is C_2 and major axis coincides with the axis of symmetry of C_1 and whose length of minor axis is 4. Identify the correct statement(s) for C_3 .

(1) Eccentricity is $\frac{1}{2}$.

(2) Focal length is 4 .

(3) Length of latus-rectum is $2\sqrt{2}$

(4) Director circle is $x^2 + y^2 - 4x - 8 = 0$.

Q41

Let $f(x) : \mathbb{R} \rightarrow [-1, 1]$ be twice differentiable and $f^2(0) + (f'(0))^2 = 4$, then which of the following(s) is/are TRUE?

(1) There exist $\alpha, \beta \in \mathbb{R}$ where $\alpha < \beta$, such that f is one-one on the interval (α, β)

(2) There exists $c \in (0, 2)$ such that $|f'(c)| \leq 1$

(3) $\lim_{x \rightarrow \infty} f(x) = 1$

(4) There exists some x_0 such that $f(x_0) + f''(x_0) = 0$ but $f'(x_0) \neq 0$

Q42

If $f(x) = ax^2 + bx + c = 0$ have real roots and its coefficients are odd positive integers then

(1) $f(x) = 0$ always have irrational roots

Questions with Answer Keys

MathonGo

(2) $\left| f\left(\frac{p}{q}\right) \right| \geq \frac{1}{q^2}$ where $p, q \in I$

(3) if $a, c = 1$, then equation must have exactly one root α such that $[\alpha] = -1$, where $[.]$ is greatest integer function

(4) equation have rational roots

Q43

Consider the set of complex number A, B, C and S defined as

$$A = \{z : ||z + 2| - |z - 2|| = 2\}$$

$$B = \left\{ z : \arg\left(\frac{z-1}{z}\right) = \frac{\pi}{2} \right\}$$

$$C = \{z : \arg(z-1) = \pi\}$$

$$S = \left\{ z : \operatorname{Re}\left(\frac{z-1}{z+1}\right) = 0 \right\}$$

If $z_1, z_2, z_3 \in S$, then find the minimum value of $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$.

Q44

If r_1, r_2 and r_3 are three values of r which satisfy the equation $\int_0^{\pi/2} (\sin x + r \cos x)^3 dx - \frac{4r}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2$ then the value of $\frac{4}{7}(r_1^2 + r_2^2 + r_3^2)$ is equal to

Q45

Let triangle ABC ($\angle C = \frac{\pi}{2}$) is right angled triangle of area equal to 12 units lie in xy plane so that its perpendicular sides are parallel to coordinates axes and the medians through A and B lie on the lines $y = 3x + 1$ and $y = mx + 2$ ($m \in \mathbb{I}$) respectively. If the coordinates of vertex C is (a, b) then $\frac{1}{|ab|}$ is equal to

Q46

Let $A_n, (n \in \mathbb{N})$ be a matrix of order $(2n-1) \times (2n-1)$, such that $a_{ij} = 0 \forall i \neq j$ and $a_{ij} = n^2 + i + 1 - 2n \forall i = j$ where a_{ij} denotes the element of i^{th} row and j^{th} column of A_n . Let $T_n = (-1)^n \times$ (sum of all the elements of A_n). Find the value of $\left\lceil \frac{\sum_{n=1}^{102} T_n}{520200} \right\rceil$, greatest integer function).

Q47

Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(x) + e^{f(x)} = \frac{2}{x} - \ln x - 1$. Find the number of integers in the range of x satisfying the inequality $f(2x^2 + 1) - f(x^2 + 5) \geq f(1), x > 0$.

Q48

Paragraph:

Let $f(x)$ be a polynomial with positive leading coefficient satisfying $f(0) = 0$ & $f(f(x)) = x \int_0^x f(t) dt \forall x \in \mathbb{R}$.

Question:

Two perpendicular tangents to the curve $y = f(x)$ will intersect on the curve/line

Questions with Answer Keys

MathonGo

(1) $x^2 + y^2 = 3$

(2) $y = \frac{1}{4\sqrt{3}}$

(3) $y = -\frac{\sqrt{3}}{4}$

(4) $y^2 = \frac{3x}{-16}$

Q49

Paragraph:

Let $f(x)$ be a polynomial with positive leading coefficient satisfying $f(0) = 0$ & $f(f(x)) = x \int_0^x f(t)dt \forall x \in \mathbb{R}$.

Question:

If a line having slope 1 and passing through $(\alpha, 0)$ neither cuts nor touches the curve $y = f(x)$, then the value of α must lie in the interval.

(1) $(0, \infty)$

(2) $\left(\frac{\sqrt{3}}{4}, \infty\right)$

(3) $\left(\frac{3-\sqrt{3}}{4}, \infty\right)$

(4) $\left(-\frac{3}{4}, \infty\right)$

Q50

Paragraph:

A and B are playing a badminton match with the agreement that winner of each set will get 1 point and the loser 0 point. The match ends as soon as one of them is ahead by 2 points or number of sets reaches six. It is supposed that the probabilities of A and B winning a set are $\frac{2}{3}$ and $\frac{1}{3}$ respectively and each set is independent.

Let X_i denotes the event that atleast i sets are played and Y and Z denotes the event that match has won by A and B respectively.

Question:

Identify incorrect option -

(1) $P\left(\frac{Y}{X_5}\right) = \frac{4}{9}$

(2) $P\left(\frac{Z}{X_4}\right) = \frac{13}{81}$

(3) $P(X_{2k-1}) = P(X_{2k}) \forall k \in \{1, 2, 3\}$

(4) $P\left(\frac{z}{x_1}\right) = \frac{64}{729}$

Q51

Questions with Answer Keys

MathonGo

Paragraph:

A and B are playing a badminton match with the agreement that winner of each set will get 1 point and the loser 0 point. The match ends as soon as one of them is ahead by 2 points or number of sets reaches six. It is supposed that the probabilities of A and B winning a set are $\frac{2}{3}$ and $\frac{1}{3}$ respectively and each set is independent.

Let X_i denotes the event that atleast i sets are played and Y and Z denotes the event that match has won by A and B respectively.

Question:

If it is known that A has won the third set then probability that A will win the match is equal to

(1) $\frac{4}{9}$

(2) $\frac{13}{18}$

(3) $\frac{22}{27}$

(4) $\frac{76}{81}$

ANSWER KEYS

1. (4)	2. (1, 4)	3. (3)	4. (2, 4)	5. (1, 3, 4)	6. (1, 4)
7. (2)	8. (1, 4)	9. (33)	10. (8.33)	11. (2.33)	12. (50)
13. (1)	14. (3)	15. (4)	16. (2)	17. (1)	18. (3)
19. (4)	20. (3)	21. (1, 3)	22. (1, 2, 3, 4)	23. (2, 3, 4)	24. (2, 3, 4)
25. (1, 3, 4)	26. (0.25)	27. (8)	28. (4)	29. (9)	30. (5)
31. (1)	32. (2)	33. (3)	34. (4)	35. (3)	36. (3)
37. (1)	38. (1, 2, 3)	39. (1, 2)	40. (2, 3, 4)	41. (1, 2, 4)	42. (1, 2, 3)
43. (3)	44. (3)	45. (0.54)	46. (2)	47. (2)	48. (3)
49. (2)	50. (4)	51. (3)			